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|              |     | Rev.  | 1          |

## OPERATING AND MAINTENANCE INSTRUCTION

|                                                                                                                                                         |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>CLIENT ADDRESS</b><br>PT Paiton Operation & Maintenance Indonesia<br>Jl. Raya Surabaya – Situbondo Km 141<br>Paiton, Probolinggo<br>Jawa Timur 67291 |                                        |
| <b>CONTACT NAME/PHONE No./E-MAIL</b>                                                                                                                    |                                        |
| <b>PROJECT</b><br>Unit 3                                                                                                                                | <b>DURAG ORDER NO.</b>                 |
| <b>PURCHASE ORDER</b>                                                                                                                                   | <b>ORDER NUMBER – LINE ITEM NUMBER</b> |
| <b>EQUIPMENT DESCRIPTION</b><br>HM-1400 TRX 2 Total Mercury Analyzer                                                                                    | <b>SERIAL NUMBER</b>                   |

|                      |                    |                    |
|----------------------|--------------------|--------------------|
| <b>ISSUE PURPOSE</b> | <b>PREPARED BY</b> | <b>APPROVED BY</b> |
| For O&M              | Wisnu              |                    |

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## 1. INTRODUCTION

### 1.1 Purpose

The purpose of this document is to provide the basic information required to operate and maintain the DURAG HM-1400 TRX2 analyzer system for Mercury Continuous Emission Monitoring (CEMS) delivered as part of the PT Paiton Operation & Maintenance Indonesia (POMI) Power plant. The manual must be used with assistance of other documents and drawings where such references are mentioned.

This manual is written as a generic document for three different installations of HM-1400 TRX2 CEMS.

### 1.2 Contact Details

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## 2. MAIN DATA

### 2.1 Scope of Supply

The purpose of this section is to ensure and identify that all of the instrumentation and equipment listed is checked and identified to confirm delivery. Any discrepancies should be noted on the comments section.

| Pos. O/C | Qty | Type                                                                       | ITEM No. | Remark |
|----------|-----|----------------------------------------------------------------------------|----------|--------|
| 100      | 1   | HM-1400 TRX 2xx MAN-EN                                                     | 4009352  |        |
| 110      | 1   | HM-1400 TRX 2EC-230-A1LDJ Total Mercury (Hg) analyzer                      | 4020469  |        |
| 120      | 1   | HM-1400 TRX 2 WT chemicals set startup                                     | 4024953  |        |
| 130      | 1   | HM-1400 S P-C-230-E sample probe with heated filter                        | 4020391  |        |
| 140      | 1   | HM-1400 E DN65-D76-L240-E - welding pipe with flange                       | 4020250  |        |
| 150      | 1   | HM-1400 S T-150-E-H heated sample tube                                     | 1700472  |        |
| 160      | 1   | HM-1400 PL-C-230-70-H - heated sample line                                 | 4029101  |        |
| 170      | 1   | Silicone connection cable H05SS-F                                          | 4014414  |        |
| 180      | 1   | HM-1400 WT LSG 0,2ppm Hg <sup>2+</sup> HCl, pH 2 1000 ml standard solution | 1702407  |        |
| 190      | 1   | HM-1400 SP sample probe filter cartridge                                   | 4022008  |        |
| 200      | 2   | HM-1400 SP maintenance set 6M                                              | 4022026  |        |
| 210      | 1   | HM-1400 SP maintenance set 12M                                             | 4022053  |        |

Table 2.1 Scope of supply

|              |                |       |            |
|--------------|----------------|-------|------------|
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## 2.2 Illustrated Part List

The table below lists main parts with reference to corresponding nameplates mounted on the analyzer system. Pictures are added for easy identification of items where relevant.

| No | Item                                                  | Picture                                                                             | Remark |
|----|-------------------------------------------------------|-------------------------------------------------------------------------------------|--------|
| 1  | HM-1400 S T-150-E-H heated sample tube                |    |        |
| 2  | HM-1400 S P-C-230-E sample probe with heated filter   |    |        |
| 3  | HM-1400 E DN65-D76-L240-E - welding pipe with flange  |    |        |
| 4  | HM-1400 PL-C-230-70-H - heated sample line            |   |        |
| 5  | HM-1400 TRX 2IC-230-A1LFM Total Mercury (Hg) analyzer |  |        |
| 6  | Silicone connection cable H05SS-F                     |  |        |

Table 2.2 Field items

| No | Item                                                                  | Picture                                                                             | Remark |
|----|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|
| 1  | Integrated HgCl <sub>2</sub> Gas Generator & Thermo Catalytic Reactor |  |        |

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|   |                                          |                                                                                     |  |
|---|------------------------------------------|-------------------------------------------------------------------------------------|--|
| 2 | Cooler                                   |    |  |
| 3 | Condensate pumps                         |    |  |
| 4 | Photometer                               |    |  |
| 5 | Hg trap filter                           |   |  |
| 6 | Sample / membrane pump                   |  |  |
| 7 | Prefilter & condensate blocking membrane |  |  |
| 8 | Zero gas filter (cartridge)              |  |  |
| 9 | Calibration gas (standard solution)      |  |  |

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|    |                                          |                                                                                     |  |
|----|------------------------------------------|-------------------------------------------------------------------------------------|--|
| 10 | Fuses                                    |    |  |
| 11 | PLC                                      |    |  |
| 12 | Terminal for AO, DI & DO                 |    |  |
| 13 | Control unit with USB stick / flash disk |    |  |
| 14 | Main switch                              |    |  |
| 15 | Air conditioning unit                    |  |  |

Table 2.3 Items in the analyzer rack

| No | Item                                                                                                                                                                                                               | Picture                                                                              | Remark                                                                                                        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|    |                                                                                                                                                                                                                    |  |                                                                                                               |
| 1  | <p>(1) 1x zero gas filter filled</p> <p>(2) 2x prefilter for particles and aerosols</p> <p>(3) 0,2m hose DI4,8x1,6-DA8-Verderprene for mounting of (1), (2) and (6)</p> <p>(6) 1x condensate blocking membrane</p> |   | <p>(1) exchange every six month</p> <p>(2) exchange every three month</p> <p>(6) exchange every six month</p> |

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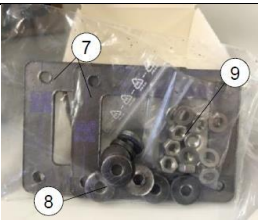
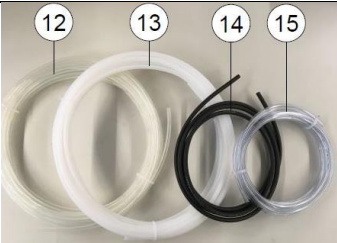
|   |                                                                                                                                                          |                                                                                    |                                                                                      |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 2 | (7) 2x sealing reduction vessel<br>100x80x2mm<br><br>(8) 16x sealing ring reduction<br>vessel $\varnothing 18/6 \times 3$ mm<br><br>(9) nuts and washers |   | (9) exchange on demand                                                               |
| 3 | 50g glass wool for Hg-trap at<br>photometer                                                                                                              |   |                                                                                      |
| 4 | 2x 60g iodine charcoal 5% for<br>Hg-trap at photometer                                                                                                   |   | exchange every 3 (three)<br>months                                                   |
| 5 | 4x 700ml solid mix for reactor<br>for conversion of $Hg^x$ into $Hg^0$                                                                                   |   | exchange the content of<br>both reduction vessels every<br>three months or on demand |
| 6 | 4x zip lock bag and label for<br>disposal of used solid mix                                                                                              |  |                                                                                      |

Table 2.4 Maintenance set for 6 months

| No | Item                                                                                                                                                                                                                                                                                                                                         | Picture                                                                                                                                                                  | Remark                               |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|    |                                                                                                                                                                                                                                                                                                                                              |   |                                      |
| 1  | (10) 1x filter mat outlet filter<br>(11) 2x filter mat filter fan                                                                                                                                                                                                                                                                            |                                                                                       | exchange of all parts once a<br>year |
| 2  | Hoses:<br>(12) 10m DI4x1-DA6 ETFE for<br>gas path starting at valves<br>( $T < 100^\circ C$ )<br>(13) 10m DI4x1-DA6 PTFE for<br>gas path in hot areas, i.e.<br>reactor ( $T > 100^\circ C$ )<br>(14) 2m DI3,2x1,6-DA6,4-Viton<br>for photometer hosing<br>(15) 3,5m DI4x1-DA6 PVC for gas<br>path behind vacuum pump and<br>condensate drain |                                                                                       | exchange on demand                   |

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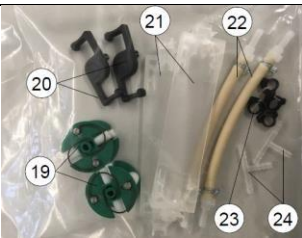
|   |                                                                                                                                                                                                      |                                                                                   |                                                 |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------|
| 3 | Wear and tear parts for maintaining of vacuum pump :<br>(16) 2x valve plate<br>(17) 1x structural membrane<br>(18) 0,3m DI4,8x1,6-DA8-Verderprene                                                    |  | exchange of all parts once a year               |
| 4 | Wear and tear parts for maintaining both condensate pumps:<br>(19) 2x cartridge with rolls<br>(20) 2x pump clip<br>(21) 2x press clamp<br>(22) 2x pump hose<br>(23) 4x hose clamp<br>(24) 2x T-piece |  | exchange of all parts on both pumps once a year |
| 5 | Fittings for gas path                                                                                                                                                                                |  | exchange on demand                              |

Table 2.5 Maintenance set for 12 months

## 2.3 Technical Specifications

|                                       |                                                                                                                                                                     |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimensions (h x w x d)                | : 1700 x 1270 x 500 mm                                                                                                                                              |
| Weight                                | : 220kg (+22kg AC unit)                                                                                                                                             |
| Material                              | : Sheet steel cabinet, painted                                                                                                                                      |
| Power supply                          | : 230/400 V 3x25A, N, PE, 50 Hz, max. 10 kVA                                                                                                                        |
| Energy demand at continuous operation | : 0.5 kWh/h (corresponds to 4500 kWh p. a.)                                                                                                                         |
| IP class (IEC 60529)                  | : IP54                                                                                                                                                              |
| Sample probe                          | : sample probe with inside heated filter element and inlet for heated test gas, PT100 temperature sensor, controlled by HM-1400 TRX 2, 230V L, N, PE 50/60Hz, 650VA |
| Sample tube                           | : heated, length 1000mm, power 230VAC L,N,PE 1200W / PT 100 supplied by HM-1400 TRX 2                                                                               |
| Sample line                           | temperature controlled 180° -200°C heated sample line, IP 65, power supply 230V L,N,PE, 50-60 Hz, 100 W/m                                                           |
| Measured components                   | : Hg total                                                                                                                                                          |
| Certified measuring ranges            | : 0...15 µg/m <sup>3</sup> , 0...45 µg/m <sup>3</sup> , 0...75 µg/m <sup>3</sup>                                                                                    |
| Analogue outputs                      | : 3x 4...20 mA, max. 500 Ohm, capable of being parameterized                                                                                                        |
| Digital outputs                       | : 9x contact NC/NO, function capable of being parameterized                                                                                                         |
| Digital inputs                        | : 8x status input, function capable of being parameterized                                                                                                          |
| Instrument air supply                 | : input pressure 6...8 bar, max. 500 l/h (in calibration gas operation)                                                                                             |
| Calibration gas                       | : HM-1400 WT LSG 0,2ppm Hg2+ HCl, pH 2 1000 ml standard solution                                                                                                    |

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## 2.4 General safety rules

### 2.4.1 General safety information

Only operate the product in perfect condition and in accordance with the operating manual. Perceptible changes from normal operation indicate functional impairments and should be taken seriously.

#### *Malfunction detection*

Look out for:

- Formation of smoke or unusual odors
- Unusual noises caused by and occurring during operation of the device (e.g. including the fan)
- Unusual vibrations
- Unusually high surface-temperatures of system components
- Unexpected changes in power consumption
- Triggering of safety monitoring devices
- Unusual and significant fluctuations or shifts in the measurement results
- Device warning and fault messages

#### *Safety information*

As a general rule, the following safety information applies for the described product:

- When preparing and performing work:

Follow the statutory regulations valid for the plant and the corresponding technical rules.

Observe national safety and accident prevention regulations.

- Maintenance work and repairs not described in this manual, must not be performed without prior arrangement with the manufacturer.

### 2.4.2 Device-specific hazards

The following section describes residual risks determined by the risk assessment.

Follow the safety instructions listed here and the warning information in the subsequent chapters of this manual in order to re-duce health hazards and prevent dangerous situations. The personnel assigned to carry out installation, commissioning and maintenance must be thoroughly familiar with all potential hazards and repair measures as set out in this manual.

#### ***Hazards due to electrical equipment***

The device is operated with electric current. There is an immediate risk to life when touching parts carrying live voltage. Damage to the insulation or individual components can present a risk to life. Work on electrical systems must only be performed by specialized electricians. Before opening the housing or removing the touch guard, disconnect the device. Check that there is no voltage and take measures to protect against unauthorized re-activation.

Where there are extreme fluctuations of temperature and moisture, condensation can cause moisture to build up within the device. This can cause an electrical short circuit. After transportation of the device, do not put it into operation until the device, including its internal parts, has acclimatized to the ambient temperature.

Only connect the HM-1400 TRX 2 to the supply voltage set out on the type label.

Only switch on the operating voltage once installation is complete. Only switch on the device once installation is complete.

Steps must be taken to protect the electronic components against electrostatic discharge (ESD protection) for all work on the open device. To prevent static charge of the human body, use a device such as a personnel earthing system.

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### ***Hazard due to hot, aggressive, explosive gases or to high pressure of gas to be measured***

The HM-1400 TRX 2 is connected to the flue gas duct via the sampling probe and the exhaust gas duct. In the case of systems with a low hazard potential (ambient pressure, low temperatures, no health hazard), mounting / removal can be carried out during system operation or when replacing connecting cables. The applicable provisions and safety regulations for the system must be complied with and any necessary and appropriate protective measures taken.

The HM-1400 TRX 2 is not suitable for use in explosive gas mixtures!

### ***Hazards due to chemicals***

Chemicals always pose special risks. They must therefore only be handled with the necessary care and by appropriately trained personnel.

The technique for measuring the mercury concentration is to pass the gas through an iodized activated carbon filter (between the measurement and reference cell of the photometer). The mercury contained in the gas being measured bonds to the carbon and is thus trapped in the filter.

A mercury chloride solution ( $\text{HgCl}_2$ ) is used for the  $\text{Hg}^{n+}$  gas generator integrated in the device. Empty containers must always be replaced by full, fresh containers. Topping up empty containers is not permissible! In addition to damage being caused as a result of spilt acidic solution in the device cabinet, changes can also occur to the available mercury content in topped-up containers.

Observe the information in the relevant safety data sheet!

Used chemicals that must be disposed of after a maximum of 12 months as part of device maintenance:

- Contents of the selective filter (iodine-active carbon contaminated with Hg)
- Mercury chloride solution
- Any collected condensate

The measuring device operating company is responsible for the handling, transportation and necessary disposal of the chemicals. All used chemicals must be observed.

The latest versions of the material safety data sheets included with the chemicals are authoritative. All persons who work with these substances must be aware of the contents of these safety data sheets.

Take all of the suitable and necessary protective measures specified in the data sheets. Also note any specified first-aid measures.

### ***Hazards due to hot surfaces***

Some of the components used in the HM-1400 TRX 2, as well as the sampling probe, sampling line, flange and flange piece external components, may be hot or may become hot during operation. There is a risk of burns. Observe the relevant warning symbols on the device components!

### ***Hazards during transportation***

The HM-1400 TRX 2 is a heavy device. There are therefore risks due to:

#### **1. Suspended loads**

Loads may swing out and fall to the ground during lifting procedures. This can lead to serious or even fatal injuries. Do not walk underneath suspended loads, or into their swivel range. Putdown the load when you leave the work area.

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2. Heavy weight

When lifting or moving heavy loads, this can lead to back pain and injury. Do not lift heavy loads without assistance. Use suit-able lifting equipment or lifting tools.

#### 2.4.3 Personnel, qualifications

Work on the HM-1400 TRX 2 must only be carried out by the following personnel:

- Specialized personnel are those who because of their specialist training, knowledge and experience, coupled with knowledge of the applicable regulations, are in a position to perform the work assigned to them and make independent judgements of the potential hazards.
- An electrician is an individual who, because of his or her specialist training, knowledge and experience, coupled with knowledge of the applicable standards and regulations, is in a position to perform work on electrical equipment and make independent judgements of the potential hazards. Electricians are specifically trained for the working environment in which they are employed.

#### 2.4.4 Description and location of safety equipment and emergency stop mechanisms

The operating company must include safety measures for the HM-1400 TRX 2 in the safety concept for the overall system. This includes the setup as well as a description of the safety and emergency-stop equipment, including information on the location of the relevant emergency-stop switches.

#### 2.4.5 Intended use

The HM-1400 TRX 2 described in this manual is used for the on-going determination of the mercury concentration in the flue gas from incineration plants and in process gases.

The device is expressly not permitted for use in potentially explosive areas.

If the device is not used as intended, the protection provided by the device may be impaired!

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### 3. PRODUCT DESCRIPTION

The HM-1400 TRX 2 is an automatic measuring system (AMS) certified to EN 15267. It is used for the ongoing analysis of all gaseous mercury compounds in systems or process gases that are subject to compulsory monitoring.

A double-beam photometer is used to determine the concentration of atomic mercury  $\text{Hg}^0$ . To determine the total mercury content  $\text{Hg}_{\text{total}}$ , oxidized mercury  $\text{Hg}^{\text{n}+}$  is reduced to  $\text{Hg}^0$  beforehand. This is performed in a thermo catalytic reactor.

The HM-1400 TRX 2 also contains the following modules for monitoring the system:

- Zero point monitoring
- Reference point monitoring
- Linearity monitoring
- System leak-tightness monitoring

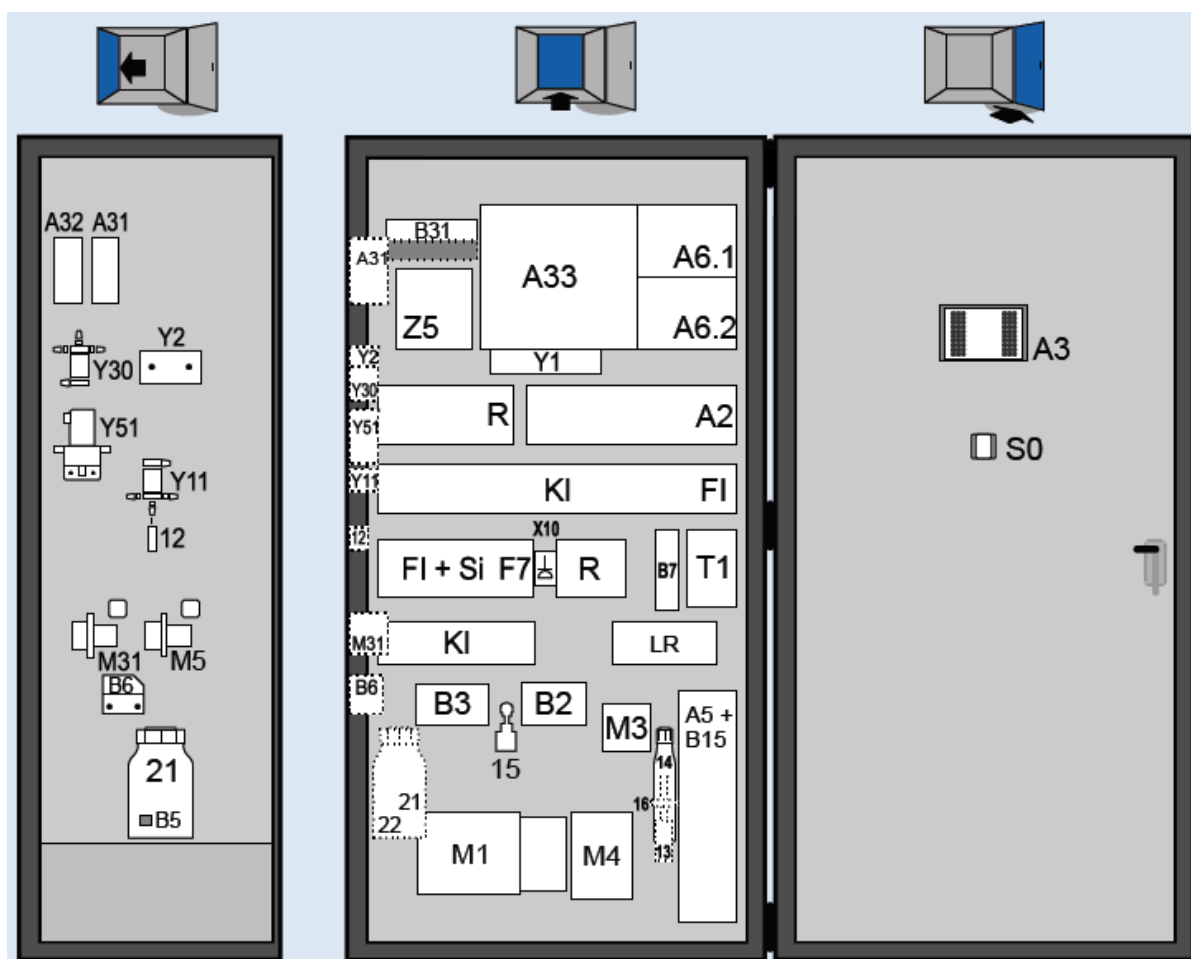


Fig. 3.1: Overview of components, designation of measurement instrument parts

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|     |                                                                           |     |                                                 |
|-----|---------------------------------------------------------------------------|-----|-------------------------------------------------|
| 12  | Hg <sup>0</sup> adsorbent (zero gas generation)                           | F7  | Fuse for reactor heating                        |
| 13  | Pre-filter for particles and aerosols                                     | FI  | RCD (residual-current device)                   |
| 14  | Hg <sup>0</sup> adsorbent (photometer)                                    | KI  | Terminals                                       |
| 15  | Needle valve for volume flow setting                                      | LR  | Overload relay                                  |
| 16  | Condensate blocking membrane                                              | M1  | Cooler                                          |
| 21  | HgCl <sub>2</sub> solution reservoir (gas generator)                      | M2  | Fan                                             |
| A2  | PLC (Programmable Logic Controller)                                       | M3  | Sample gas pump                                 |
| A3  | Control unit                                                              | M4  | Condensate pumps                                |
| A5  | Hg <sup>0</sup> photometer                                                | M31 | Peristaltic pump for HgCl <sub>2</sub> solution |
| A6  | Thermo-catalytic reactor                                                  | R   | Relay                                           |
| A31 | Mass flow controller for gas generator                                    | S0  | Main switch                                     |
| A33 | HgCl <sub>2</sub> evaporator                                              | Si  | Fuse(s)                                         |
| B2  | Differential pressure sensor                                              | T1  | Power supply unit                               |
| B3  | Absolute pressure sensor                                                  | X10 | Socket for ext. Test gas generator              |
| B5  | Level gauge for monitoring the HgCl <sub>2</sub> solution (gas generator) | Y1  | 3-way valve (zero gas/sample gas)               |
| B6  | Mass flow meter for HgCl <sub>2</sub> solution                            | Y2  | Selection valve for reduction vessel A6.1/ A6.2 |
| B7  | Safety temperature limiter for the reactor                                | Y11 | Leak test/zero gas valve                        |
| B15 | Sample gas temperature sensor                                             | Z5  | Mains filter                                    |

Table 3.1: Overview of instrument components

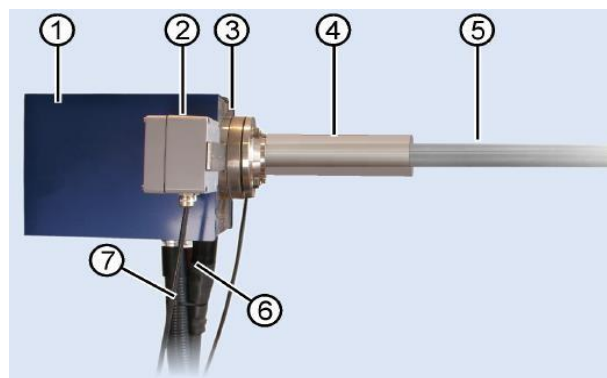


Fig. 3.2: Overview of components, designation of sampling probe

|   |                                         |   |                                                                                       |
|---|-----------------------------------------|---|---------------------------------------------------------------------------------------|
| 1 | Sampling probe housing cover            | 5 | Sampling tube                                                                         |
| 2 | Electric terminal box                   | 6 | Gas line from external test gas generator<br>(option - not installed on this variant) |
| 3 | Adapter flange for heated sampling tube | 7 | Sample gas line                                                                       |
| 4 | Welded-in tube with installation flange |   |                                                                                       |

Table 3.2: Overview of sampling probe components

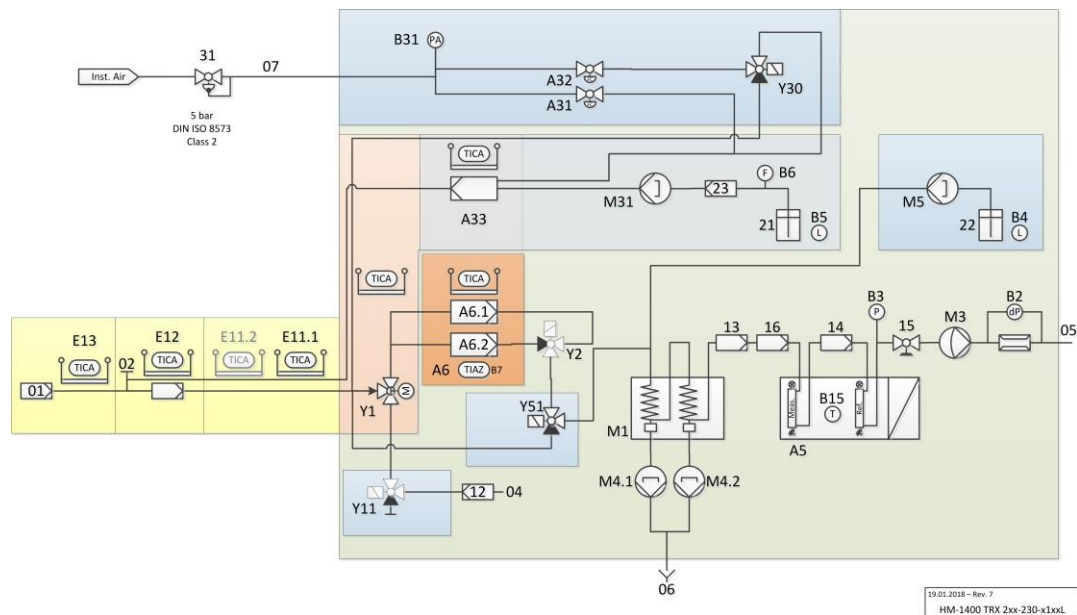


Fig. 3.3: Substance flow plan HM-1400 TRX 2

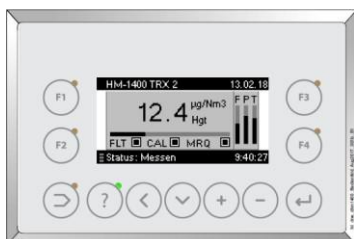
|      |                                                 |       |                                                                           |
|------|-------------------------------------------------|-------|---------------------------------------------------------------------------|
| 01   | Gas ingress into stack                          | B2    | Differential pressure sensor                                              |
| 02   | Test connection (test gas application)          | B3    | Absolute pressure sensor                                                  |
| 04   | Zero gas                                        | B4    | Level gauge for monitoring the KOH solution (speciation) ①                |
| 05   | Exhaust gas                                     | B5    | Level gauge for monitoring the HgCl <sub>2</sub> solution (gas generator) |
| 06   | Condensate                                      | B6    | Mass flow meter for HgCl <sub>2</sub> solution                            |
| 07   | Instrument air                                  | B7    | Safety temperature limiter                                                |
| 12   | Hg <sup>0</sup> adsorbent (zero gas generation) | B15   | Temperature sensor for the sample gas in the photometer                   |
| 13   | Pre-filter for particles and aerosols           | B31   | Pressure sensor for instrument air inlet                                  |
| 14   | Hg <sup>0</sup> adsorbent (photometer)          | E11.1 | Sample gas line (heated)                                                  |
| 15   | Needle valve for volume flow setting            | E11.2 | Extension for the heated sample gas line ①                                |
| 16   | Condensate blocking membrane                    | E12   | Sampling probe                                                            |
| 21   | Reservoir (HgCl <sub>2</sub> gas generator)     | E13   | Sampling tube                                                             |
| 22   | KOH solution reservoir (speciation) ①           | M1    | Cooler                                                                    |
| 23   | Damping for mass flow meter                     | M3    | Sample gas pump                                                           |
| 31   | Pressure switch                                 | M4.1  | Condensate pump                                                           |
| A5   | Hg <sup>0</sup> photometer                      | M4.2  | Condensate pump                                                           |
| A6   | Thermocatalytic reactor                         | M5    | Peristaltic pump for KOH solution ①                                       |
| A6.1 | Reduction vessel 1                              | M31   | Peristaltic pump for HgCl <sub>2</sub> solution                           |
| A6.2 | Reduction vessel 2                              | Y1    | 3-way valve (zero gas/sample gas)                                         |
| A31  | Mass flow controller for gas generator          | Y2    | Selection valve for reduction vessel A6.1/A6.2                            |
| A32  | Mass flow controller for dilution ①             | Y11   | Leak test/zero gas valve                                                  |
| A33  | HgCl <sub>2</sub> evaporator                    | Y30   | Switchover valve for dilution ①                                           |
| ①    | Option (not installed on this variant)          | Y51   | Valve for speciation ①                                                    |

Table 3.3: Overview of HM-1400 TRX2 components

For theoretical basis of this system, please refer to the Manual of HM-1400 TRX 2.

## 4. OPERATION

### 4.1 The Terminal



The HM-1400 TRX 2 is operated via the control panel in the door of the device.

Fig. 4.1 Control panel

| Key Functions                                                                       |                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Edit keys                                                                           |                                                                                                                                                                                                                                                                           |
|    | <b>The Plus key</b> is used to change input values. The selected value will increase by 1 when the key is pressed. Keeping the key held down will repeat the function. The speed at which the function repeats will increase automatically as the key is kept held down.  |
|    | <b>The Minus key</b> is used to change input values. The selected value will decrease by 1 when the key is pressed. Keeping the key held down will repeat the function. The speed at which the function repeats will increase automatically as the key is kept held down. |
| Control keys                                                                        |                                                                                                                                                                                                                                                                           |
|   | <b>The Cursor left key</b> directly selects neighboring screens or messages, ends the operating mode or moves the cursor one character to the left within a variable.                                                                                                     |
|  | <b>The Cursor down key</b> directly selects neighboring screens or messages, switches to the messages list during operation or moves the cursor downwards to the next editable variable.                                                                                  |
| Special keys                                                                        |                                                                                                                                                                                                                                                                           |
|  | <b>The Help key</b> displays the current help text or fault messages. If the help key LED is flashing, this indicates that messages are waiting to be viewed.                                                                                                             |
|  | <b>The Release data key</b> takes you from the menu to the editor. If it is pressed whilst in editor mode, then editor mode will be exited again.                                                                                                                         |
|  | <b>The Enter key</b> accepts the current data then switches over to the next field (use the Release Data key (see above) when you have finished inputting all fields).                                                                                                    |
|  | <b>Function key F1 (Home)</b> is used to return to the standard operating screen (large display of current concentration). You can reach this screen from any menu.                                                                                                       |
|  | <b>Function key F2 (Back)</b> exits a menu or screen and takes you to the previous menu or screen.                                                                                                                                                                        |
|  | <b>Function key F3 (Event list)</b> takes you to the event list (messages) from any menu.                                                                                                                                                                                 |
|  | <b>Function key F4 ("ENTER")</b> executes the highlighted function in the current screen.                                                                                                                                                                                 |

Table 4.1 Key functions of control panel

For detail information, please refer to the Manual of HM-1400 TRX 2.

|                                                                                  |     |       |            |
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## 4.2 Basic Rule Operation

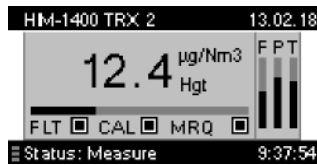


Fig. 4.2 Main screen

Once the device has started up and the initialisation process is complete, the current measuring instrument status is shown on the main screen (Fig. 4.2) on the operating panel. The display will start by showing the current measured value for the total mercury content  $Hg_{total}$ .

The Down and Left keys can then be used to scroll through additional displays at the main menu level.

The measured values will be displayed in the two measurement ranges. The range can be adjusted as per requirement.



Fig. 4.3 Maintenance switch

Pressing the Down key again will open the maintenance switch. The F4 key can be used to select whether the maintenance output is active/inactive or automatically controlled:

- Central position (normal operation): The device is in normal operation and the CAL output is controlled by the PLC depending on the status.
- Maintenance (maintenance mode): The device is in maintenance mode and the CAL output is always set. All automatic internal processes such as zero adjustment, reference runs, speciation cycles or leak tests are deactivated.
- Operation (forced operating mode): An external maintenance input is blocked and the device reports operational readiness regardless of the external maintenance input. The maintenance output is still set in the event of zero adjustment, reference runs etc.

There are a number of ways of selecting a specific menu screen from a particular operating situation.

- Pressing the F1 key always takes you immediately back to the main screen (Fig. 4.2) as the defined starting point. You can then start the menu system from this point by pressing F4. The Down and Left keys can be used to select one of the 4 additional main menus. Pressing the F4 key takes you to the selected sub-menu. The other sub-menus are accessed in the same way. The F2 key can be used to access the higher-level menu.
- Additional help text is displayed in the right-hand section of the display in some menus, in order to help you make a decision on your next operating step.

Abbreviations are often used when working in the display. For the meaning of these abbreviations, please refer to Manual of HM-1400 TRX 2 (table 6.2).

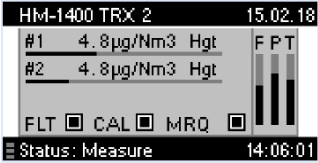
## 4.3 Measured Value & System Structure

There are five main menus for the operation of the HM-1400 TRX 2:

- Menu branch 0 contains various display screens including messages and the start screen
- Menu branch 1 contains display screens for exchanging signals with the measuring station
- Menu branch 2 contains display screens for controlling the device ("switches")
- Menu branch 3 contains display screens for settings and parameter configuration
- Menu branch 4 contains display screens for querying internal parameters and statuses

Once the device has been switched on and the software initialization is complete, the display moves to the main screen in menu branch 0 (see Fig. 4.2) and stays there.

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|                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Fig. 4.4 Maintenance switch</p> | In this screen, the Down and Left keys can be used to move between the main screen and the measured value display in the top display level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Upper black info bar:                                                                                                | To the left: Device type. To the right: Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Lower black status bar:                                                                                              | <ul style="list-style-type: none"> <li>- 3 overlying bars symbolizing the status messages FLT, CAL and MRQ.</li> <li>- The current operating status is displayed next to these bars</li> <li>- On the right is the current time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| In the middle                                                                                                        | <ul style="list-style-type: none"> <li>- Display of the total mercury concentration <math>Hg_{total}</math> in <math>\mu g/Nm^3</math>, and as a scaled bar for the measurement range underneath.</li> <li>- The system variables of volume flow (F), pressure (P) and sample gas temperature (T) are displayed on the right in the form of bar diagrams in relationship to the specified limits.</li> <li>- The statuses of faults (FLT), maintenance or calibration (CAL) and maintenance requirements (MRQ) are shown in a bar underneath the measured values, depending on the parameterization (white field = active; black field = inactive).</li> </ul> |

#### 4.4 System Message

The message screen is menu 0. You can always access this screen via F3.

You can use the F3 and F4 keys in the message level to move forwards and backwards through individual message groups to view the individual messages saved there. Individual messages can be fault messages, status messages and result messages, which can be summarized as follows:

- Logbook: Contains messages combined from groups FLT, CAL and MRQ
- FLT: currently queued fault messages
- CAL: Standby (internal), zero/reference test messages
- MRQ: Maintenance requirement, gas temperature messages

Use the Down and Left keys to scroll through the messages within the selected message group.

Press the F1 key to leave the message level. The last screen will appear again on the display. The messages will not be deleted.

During active recording, all messages can also be stored onto a USB stick in the form of a log file

*For detail of all menus, please refer to the Manual of HM-1400 TRX 2 section 6-6.*

#### 4.5 Zero adjustment

The measurement of the mercury concentration in the photometer is based on the attenuation of the light intensity. The intensity without mercury in the cell is always used as a reference value. This changes over time due to impurities in the cell, discoloration of the glass parts as a result of interaction with the UV light and wear of the lamp and photo sensors.

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In order to ensure that the measurement signal remains stable and that the measured values do not move upwards or into the negative range, the zero position is determined on a regular basis.

#### 4.6 Verification of the measuring system

The HM-1400 TRX 2 must be calibrated before the start of regular measurements, e.g. during initial commissioning or following major maintenance work. An adjustment or check, already scheduled automatically at regular intervals.

For the regular adjustment/checking of the HM-1400 TRX 2,  $\text{HgCl}_2$  is used with the internal gas generator (*HM-1400 WT LSG 0,2ppm  $\text{Hg}^{2+}$  HCl, pH 2 1000 ml standard solution*). The internal gas generator generates oxidized mercury  $\text{Hg}^{n+}$  from the  $\text{HgCl}_2$ .

*For detail of adjusting with the internal gas generator, please refer to the Manual of HM-1400 TRX 2 section 7-3.*

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## 5. MAINTENANCE

Maintenance of the HM-1400 TRX 2 and the optional components must only be carried out by the manufacturer or by personnel who have verifiably undergone training by the manufacturer. The maintenance work and the necessary tools are described in detail in the maintenance instructions for the HM-1400 TRX 2. This document is only available to those people who have completed a training session put on by the manufacturer.

The maintenance intervals are defined by the operating company, depending on the particular system conditions. If the HM-1400 TRX 2 indicates a maintenance requirement or faults, then decisions on the time frame and necessary measures must be made on a case-by-case basis.

The first step in any maintenance work is to establish the current condition of the device. The necessary measures for the HM-1400 TRX 2, the sampling system and any internal and external gas generators are then defined. The test and repair requirements are then entered in the maintenance plan and processed. Each maintenance task ends with a final check of the HM-1400 TRX 2.

### 5.1 Maintenance Schedule

Maintenance routines must be performed regularly as stated in the following table to ensure optimal performance and a long operational life of the system and its components.

The preventive maintenance program shall, by defined time intervals, guide the technician through an inspection, repair and replacement program. This will enable the system to be operational, to reduce the number of expected alarms, to eliminate the number of unexpected alarms and to conduct correct measurements.

The proposed maintenance schedule must be accommodated to authoritative legislation and actual operational conditions. The customer should thus work out a local maintenance plan.

The following table provides an overview of the necessary work steps with the corresponding time intervals.

| No | Unit                                | Maintenance Work                                                                                               | Responsibility |
|----|-------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------|
| 1  | Cabinet and component exterior      | Clean all surfaces                                                                                             | End-User       |
| 2  | All components                      | Inspect visually for deposits, damages, corrosion etc.                                                         | End-User       |
| 3  | All instruments                     | Check status in menu 41...46<br>Check for alarms and fault messages.<br>Inspect information messages and logs. | End-User       |
| 4  | Heated sample line                  | Check status in menu 41...46<br>Check status temperature by hand.<br>Sample tubes should be noticeably warm.   | End-User       |
| 5  | Calibration gas (standard solution) | Check that there is sufficient HgCl <sub>2</sub> in the bottle.                                                | End-User       |
| 6  | Air conditioning unit               | Clean unit exterior<br>Check that air inlet and outlet vents are free from sand, dust or other foreign matters | End-User       |

Table 5.1 Daily routine check

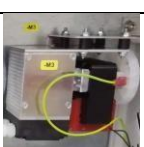
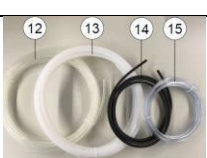
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| No | Maintenance work                                                                                                         |                                                | Months |   |   |    | Responsibility |
|----|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------|---|---|----|----------------|
|    |                                                                                                                          |                                                | 3      | 6 | 9 | 12 |                |
| 1  | Determine the current status of the overall measuring system                                                             | Menu 41 .... 46                                | X      | X | X | X  | Specialist     |
| 2  | Read out, document and check all parameters, sensor information and operating data                                       | Menu 41 .... 46                                | X      | X | X | X  | Specialist     |
| 3  | Read out, document and check the fault log and calibration messages                                                      | USB/Memory stick                               | X      | X | X | X  | Specialist     |
| 4  | Check the leak-tightness of the HM-1400 TRX 2                                                                            | Menu: 224                                      | X      | X | X | X  | Specialist     |
| 5  | Check the sampling system, including a test of the leak-tightness and the cold spots and a check of the heating systems. | Leak: menu 224<br>Temperature: menu 41 .... 46 | X      | X | X | X  | Specialist     |
| 6  | Function test of the internal gas generator including a check of the flow rate and zero gas test.                        | Zero: 32111<br>Ref#0: 242                      | X      | X | X | X  | Specialist     |
| 7  | Reference point check with Hg <sup>nt</sup>                                                                              | Manual Ref: 241....<br>Cycle: 32515            | X      | X | X | X  | Specialist     |
| 8  | Check the flow measurement                                                                                               | Menu 41 .... 46                                | X      | X | X | X  | Specialist     |
| 9  | Measurement-based check of the temperature sensors in the reactor (A6) and on the 3-way valve (Y1)                       | Menu 41...46<br>Adjustment: menu 3214          | X      | X | X | X  | Specialist     |
| 10 | Check the residual current device                                                                                        |                                                | X      | X | X | X  | Specialist     |
| 11 | Check the age of the battery in the PLC and in the display                                                               |                                                |        |   |   | X  | Specialist     |

Table 5.2 Regularly maintenance check

| No | Maintenance work                                                                        |                                                                                     | Months |   |   |    | Responsibility |
|----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|---|---|----|----------------|
|    |                                                                                         |                                                                                     | 3      | 6 | 9 | 12 |                |
| 1  | Replace particle protection filter in the sampling probe                                |  | X      | X | X | X  | Specialist     |
| 2  | Replace the particle protection filter (13) upstream from the photometer                |  | X      | X | X | X  | Specialist     |
| 3  | Replace the iodine-active carbon in the Hg <sup>0</sup> adsorber (14) on the photometer |  | X      | X | X | X  | Specialist     |
| 4  | Clean the cooling body of the gas cooler (M1)                                           |  | X      | X | X | X  | Specialist     |

|              |                |       |            |
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|    |                                                                             |                                                                                     |   |   |   |   |            |
|----|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---|---|---|---|------------|
|    |                                                                             |                                                                                     |   |   |   |   |            |
| 5  | Adjust the flow at the needle valve (15)                                    |    | X | X | X | X | Specialist |
| 6  | Carry out a photometer zero adjustment (A5)                                 | Menu 32111                                                                          | X | X | X | X | Specialist |
| 7  | Rinse the wash bottles of the gas cooler (M1) with water                    |    |   | X |   | X | Specialist |
| 8  | Replace the reactor filling (A6.1 and A6.2)*                                |    |   | X |   | X | Specialist |
| 9  | Replace the condensate blocking membrane (16)                               |    |   | X |   | X | Specialist |
| 10 | Replace the membrane of the gas pump (M3)                                   |   |   |   |   | X | Specialist |
| 11 | Maintain the condensate pump (M4.#), replace the hoses and pressure rollers |  |   |   |   | X | Specialist |
| 12 | Replace the iodine-active carbon (12) in the zero gas filter                |  |   |   |   | X | Specialist |
| 13 | Replace all hoses in the gas path from the reactor to the photometer        |  |   |   |   | X | Specialist |
| 14 | Adjust the flow                                                             |  |   |   |   | X | Specialist |
| 15 | Test the AC unit in accordance with the manufacturer's specifications       |                                                                                     |   |   |   | X | Specialist |

|              |                |       |            |
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|    |                                                                |                                                                                   |  |  |  |   |            |
|----|----------------------------------------------------------------|-----------------------------------------------------------------------------------|--|--|--|---|------------|
| 16 | Replace the hoses in the pumps (M31) for the HgCl <sub>2</sub> |  |  |  |  | X | Specialist |
| *  | <i>Or as required</i>                                          |                                                                                   |  |  |  |   |            |

Table 5.3 Regularly recurring maintenance work for 1 year

| No | Maintenance work                                                                | Time Intervals | Responsibility |
|----|---------------------------------------------------------------------------------|----------------|----------------|
| 1  | Fully replace the condensate pumps (M4.#)                                       | ≥ 2 years      | Specialist     |
| 2  | Replace and calibrate the UV lamp in the photometer (performed by manufacturer) | ≥ 2 years      | Specialist     |
| 3  | Replace the battery in the PLC and in the display                               | 5 years        | Specialist     |

Table 5.4 Regularly recurring maintenance work for more than 2 years operation

| No | Final check after all maintenance work                                                              | Responsibility |
|----|-----------------------------------------------------------------------------------------------------|----------------|
| 1  | Check the status of all temperature controllers and all analogue and digital sensors                | Specialist     |
| 2  | Check the leak-tightness                                                                            | Specialist     |
| 3  | Check the span value and linearity using an internal gas generator                                  | Specialist     |
| 4  | Test the internal gas generator                                                                     | Specialist     |
| 5  | Reset the maintenance counter, delete the fault log and set the operating data for normal operation | Specialist     |
| 6  | Check normal operation                                                                              | Specialist     |
| 7  | Create current configuration file                                                                   | Specialist     |

Table 5.5 Final check after all maintenance work

## 5.2 Procedure of replace reactor filling (A6.1 and A6.2)



### **Risk posed by hot surfaces!**

Contact with hot components may cause serious burns.

Always use suitable heat-protective equipment (such as protective gloves.)

There is only ever one reduction vessel in use during measurement operation. This means that when the first filling reaches the end of its service life, it is possible to switch to the second filling with barely any interruption of the measurement operation.

1. Open the reactor box.
2. Loose the screw reactor plate using 5mm Allen key (hex key).
3. Remove the plate using provided tool.
4. Pull the box.
5. Place the box in horizontal position to prevent activator charcoal washers being damage.
6. Wait until the box temperature cooler, around ambient temperature.
7. Remove 4 nuts.
8. Take out the plate.
9. Pour the charcoal & glass wool into zip lock bag. Mark the zip lock bag using label.
10. Replace gasket and washer behind the box with new spare part.
11. Re-fill the box using glass wool + charcoal + glass wool.
12. Close it back and secure with 4 nuts.

|              |                |       |            |
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13. Use the special tool, and the box up, place the box in to the reactor.
14. Place the plate to the reactor.
15. Tighten the screw using 5mm Allen key.
16. Close the cover of reactor boxes.

### 5.3 Procedure of replace iodine-carbon Hg trap filter

1. Make sure the unit already turn-off.
2. Loosen the tube that connected to the bottle.
3. Loosen the clip to remove the bottle.
4. Remove bottle.
5. Open the bottle cap (red).
6. Pour the charcoal & glass wool into zip lock bag. Mark the zip lock bag using label.
7. Re-fill the box with: glass wool + charcoal + glass wool.
8. Put on bottle cap.
9. Put the bottle back in its place.
10. Tighten the clip.
11. Re-install the tube

### 5.4 Planned Operational Stop

A planned operational stop is performed for example prior to scheduled maintenance or replacement of components.

1. If necessary, operate local valves or regulators to stop flow through components.
2. Allow the pressure to bleed off prior to any disassembly.
3. Maintenance of all components may now be performed. Please refer to Table 5.1 - 5.4.
4. When maintenance work is finished, restart instruments (if relevant) and reset the position of all valves or regulators being operated. Please refer to the Table 5.5.

### 5.5 Emergency Stop

An emergency stop shall be performed when disturbances of normal operation or other system malfunctions are detected.

1. Immediately shut down system power using the main switch.
2. Locate the cause of failure. Use the trouble-shooting instructions supplied by manufacturer.
3. When the fault has been rectified, restart the system.